

Executive Summary and Recommendations – Telecom Customer Churn Analysis

Project Overview

This project analyzes customer churn behavior in a telecom company using a dataset containing 7,043 customer records and 21 service-related features. The objective was to identify the major factors contributing to customer attrition and provide actionable business recommendations to improve customer retention.

Data Preparation

The dataset was first cleaned and preprocessed by handling missing values in the TotalCharges column and converting it into a numerical format suitable for analysis. Categorical variables were standardized and transformed where required to improve readability and visualization.

Overall Churn Analysis

Out of 7,043 customers analyzed:

- 5,174 customers (73.5%) remained with the company.
- 1,869 customers (26.5%) churned and discontinued the service.

This indicates that approximately 1 out of every 4 customers leaves the company, making churn reduction a significant business concern.

Key Findings

1. Contract Type is the Strongest Predictor of Churn

Customers subscribed to Month-to-Month contracts exhibit significantly higher churn rates compared to customers on One-Year and Two-Year contracts.

Insight: Customers with long-term commitments are substantially more loyal and less likely to leave the company.

2. Customer Tenure Strongly Influences Retention

Customers with lower tenure (new customers) show the highest churn behavior, while customers who have remained with the company for longer periods exhibit significantly lower churn rates.

Insight: The first few months of customer onboarding represent the highest-risk period for customer loss.

3. Senior Citizens Have Higher Churn Propensity

Although senior citizens represent a smaller proportion of the customer base, their churn percentage is considerably higher compared to non-senior customers.

Insight: Senior customers may require additional support, personalized assistance, or specialized service plans.

4. Internet Service Type Plays a Major Role

Among all internet service categories:

- Fiber Optic customers demonstrate the highest churn rate.
- DSL customers show lower churn.
- Customers without internet service exhibit the lowest churn.

Insight: High churn among Fiber Optic users may indicate pricing concerns, service quality issues, or unmet customer expectations.

5. Security and Support Services Improve Retention

Customers without the following services show significantly higher churn:

- Online Security
- Online Backup
- Device Protection
- Tech Support

Customers who subscribe to these services demonstrate stronger retention behavior.

Insight: Value-added services increase customer engagement and reduce the likelihood of customer attrition.

6. Streaming Services Have Minimal Impact

Streaming TV and Streaming Movies show relatively small differences in churn behavior between customers who subscribe and those who do not.

Insight: Entertainment services alone are not strong churn predictors.

7. Phone Service Has Limited Influence

Phone Service and Multiple Lines do not demonstrate a strong relationship with customer churn.

Insight: Core connectivity services are not primary drivers of customer attrition.

8. Payment Method Analysis

Customers using Electronic Check payments exhibit the highest churn rates among all payment methods.

Insight: Electronic Check users represent a high-risk customer segment and should be targeted through retention campaigns.

9. Monthly Charges and Churn

Customers paying higher monthly charges tend to exhibit higher churn behavior compared to customers with lower monthly bills.

Insight: Pricing sensitivity plays an important role in customer retention.

Major Churn Drivers Identified

The analysis identified the following factors as the strongest contributors to customer churn:

1. Month-to-Month Contract
2. Low Customer Tenure
3. Fiber Optic Internet Service
4. Absence of Online Security
5. Absence of Online Backup
6. Absence of Device Protection
7. Absence of Tech Support
8. Electronic Check Payment Method
9. Higher Monthly Charges
10. Senior Citizen Status

Business Recommendations

Based on the findings, the following actions are recommended:

1. Promote long-term contracts through discounts and loyalty rewards.

2. Develop retention programs specifically targeting new customers during their first few months.
3. Investigate the reasons behind high churn among Fiber Optic users.
4. Bundle security and support services into attractive packages.
5. Launch personalized retention campaigns for Electronic Check users.
6. Provide dedicated customer support for senior citizens.
7. Offer pricing incentives or customized plans for customers with high monthly charges.

Conclusion

The analysis reveals that customer churn is primarily influenced by contract type, customer tenure, internet service category, payment behavior, pricing, and the availability of support-related services. Customers on Month-to-Month contracts, those with shorter tenure, Fiber Optic subscribers, and customers lacking security or support services are considerably more likely to leave the company.

By focusing on these high-risk customer segments and implementing targeted retention strategies, the telecom company can significantly reduce churn, improve customer satisfaction, and increase long-term revenue.